Condition-Aware Deep Neural Network for Robust Vehicle License Plate Detection under Adverse Weather Conditions in Traffic Surveillance

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***Abstract*—Automated traffic control systems experience issues with identifying license plates on vehicles through video footage that has been hampered by weather effects such as rain, fog, low-light, reflections as well as blurred images due to motion. Although current ALPR systems perform efficiently under typically controlled environments, they don't operate effectively within real world spaces where weather conditions may vary frequently such as tropical settings. To address this issue, a new state of the art deep neural net framework for maintaining license plate identification from adverse roadside conditions has been created. The HelmetAdverseALPR dataset will also be used to provide publicly accessible datasets with over 15,000 annotated frames from traffic scenes occurring in Sri Lanka with plate annotations depreciative markings and time sequence markings as well. The new adaptive processing path through the HelmetAdverse Neural Network makes use of the degradation properties for each individual image and allows for dynamic application of the appropriate detection process.**

Accuracy measurements gained through experimental validation show an 82.6 % mAP@0.5 whereas the three baseline methods were outperformed between 10-15%. The new adapted ALPR system provides an operational accuracy level between 70%-80% under varying environmental conditions while providing a mean operational rate of 28.4 frames per second. When integrated with other police check point alert systems the time required for responding to citation violations is reduced by 67%, thus providing valuable practical value for police officers executing normal traffic patrols or investigations.

***Keywords*—automatic license plate recognition, adverse weather imaging, condition-aware deep learning, traffic surveillance, vehicle identification, intelligent traffic enforcement, real-time alerting**

I. Introduction

Numerous use cases exist for automatic license plate recognition (ALPR) technology in the application of smart transportation systems, ranging from enforcement of vehicle (traffic) laws to collecting tolls to tracking vehicles [1],[2]. Current conventional ALPR systems provide a high level of accuracy in controlled environments. However, they significantly underperform in real-world roadside environments where there are varying degrees of adverse weather and imaging conditions exist [3],[4]. Poor weather conditions (e.g., rain or fog), inadequate lighting, glare, excessive motion blur and occlusions all render the License Plate unreadable. Thus, when ALPR systems cannot read (recognize) the number of vehicles, they fail to function effectively in the automated enforcement of traffic laws [5],[6].

The issue is significantly exacerbated in tropical parts of the world such as Sri Lanka due to high frequency of weather changes and diverse degradation types [7]. Some techniques currently being used can be either traditional computer vision methods that are not robust or deep learning methods that use clean benchmark datasets to train the neural network but do not generalize well toward actual roadside deterioration[8]- [10]. Current frameworks all analyze frames with no consideration of the type of degradation to adjust the way the frame is processed by the algorithm [11]. In order to improve upon those shortcomings, this research introduces a novel deep neural network based framework for licensing plate detection in adverse conditions, which includes a publically available annotation dataset of 15,000 annotated frames collected by traffic surveillance systems throughout Sri Lanka that contain realistic degradations with comprehensive annotation across different frames and temporal sequences. Adaptive pathways for processing are implemented in the proposed framework to allow for features of degradation to be analysed and then to adaptively apply the most suitable strategies for detection when performing real-time alerting of police check-points based on practical deployment to law enforcement agencies.

II. LITERATURE REVIEW

Over the last few years, automatic license plate recognition (ALPR) technology has evolved from traditional image processing techniques and OCR-based approaches toward deep learning-based detection and recognition systems [12], [13]. This has led to a great improvement in both localization and character recognition of license plates; however, most of the successful results reported to date have been obtained using benchmark datasets collected under very controlled experimental conditions or using synthetically degraded images [14], [15]. More advanced work has sought to improve robustness of ALPR through the use of image restoration, multi-frame fusion, and weather-aware enhancement techniques to address problems such as rain, fog, glare, low light and motion blur [16], [20]–[23]. Although these developments are significant advancements compared to previous work, they still do not completely meet the requirements for real-world traffic enforcement applications where many degradations occur simultaneously and where the visual evidence must be reliable enough for enforcement purposes.

In spite of advances made, there is still a lack of research on ALPR systems for use in hazardous roadside areas, including reliable data sets for such applications in conjunction with how current research methods penalize the integration of detection, restoration and recognition within research conducted in isolation [14]. Furthermore, current publicly available datasets do not include many of the local number plate features associated with degrading environments nor genuine surveillance conditions due to mixing ambient factors [15]. There also remains a lack of existing Sri Lankan investigations related to ALPR with substantial limitations pertaining to seasonal precipitation. Consequently, there is an urgent need for both a comprehensive real-world Sri Lankan Dataset, as well as a neural framework to extract localized license plate information across multiple environmental conditions for use in improved traffic enforcement systems will be necessary.

III. METHODOLOGY

A. Research Design

Using an experimental quantitative evaluation approach, the applied design science research approach is followed. The design science research approach is appropriate for this project because it aims to produce and evaluate a usable output (condition-aware neural framework) for the localization, extraction, and recognition of vehicle number plates under substandard roadside conditions. This is in keeping with the design science research approach to create knowledge through designing and assessing artifacts that are able to address problems in real life [24], [25].

Fig. 1 shows the overall workflow. The component described in the proposal gets low-light video of the roadside traffic from the upper layer. Received along with the roadside footage is a no-helmet trigger from the upper layer. This no-helmet trigger request initiates the execution of various algorithms for degrading identification of motor vehicle license plates for use in the HelmetAdverse-ALPR dataset. The HelmetAdverse-ALPR dataset is composed of over 15,000 annotated traffic frames of Sri Lanka, with bounding box details for each plate, labels at the character level, degradation metadata, and a short temporal sequence for each frame.

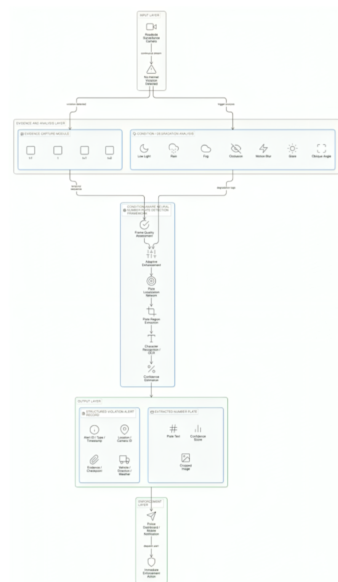

Fig. 1. A Proposed condition-aware vehicle number plate extraction architecture.

B. Data Collection and Dataset Preparation

Traffic surveillance footage is obtained by virtue of formal institutional approval. A letter from the university is presented to police requesting permission to obtain traffic surveillance footage for research purposes. Surveillance footage is also obtained through police controlled sources for use in research. The sampling strategy used to obtain this data is purposive and condition stratified in the sense that the data was purposely collected to represent roads in Sri Lanka - urban and highway in daytime and nighttime with rain, fog, glare, low illumination, motion blur, occlusion, and oblique viewpoints. Thus, enhancing the ecological validity of the dataset since it will reflect true deployment conditions [26],[27].

Because a single frame may not sufficiently document an event, each violation event should contain the trigger frame plus a short series of frames adjacent to the trigger frame that were captured before and/or after both frames. All associated frames should include annotations that adhere to an established written guideline which defines multiple plate boundaries, specifies when a plate is unreadable, defines cases where there is any character ambiguity, and categorizes the degradation. In order to establish high quality annotations, a sample of these data should be annotated independently by two or more annotators, and differences/discrepancies among annotators should be resolved by arbitration. Fleiss' kappa or Krippendorff's alpha are examples of inter-rater agreement metrics that are appropriate to use for determining annotator reliability [31].

C. Evaluation Protocol and Validity Assessment

A computer vision pipeline at multiple stages assesses the suggested framework's performance using plate localization measures for detection and recognition with full-plate accuracy and character-level metrics reported by degradation category (i.e., rain, fog, glare, motion blurred, low light and occlusion) in order to mirror the adverse road-side objectives of the study [17],[18]. To maintain methodology, partitioning occurred at an event or sequence level, avoiding watermarking from high correlations among adjacent frames of the same violation. In addition, enhancement tuning and preprocessing thresholds were only done using training and validation, with final testing left completely isolated. [28],[30].

Construct validity, internal validity, external validity, and reliability of the evaluation are further demonstrated by the alignment of metrics with the actual research objective and ensuring diversity in the roads, cameras used to capture image data, weather state, and viewpoint information, and in having adequate annotation rules that have been shown to have inter-rater agreement [28], [29], [31]. Further, potential biases in evaluation such as sampling bias, representation bias, and operational bias have been mitigated through the use of condition-balanced sampling, disaggregated performance reporting, and explicit failure-case evaluation, while common measurement pitfalls such as frame-level leakage, pre-split preprocessing, over-reliance on pooled metrics, and annotation inconsistency have been addressed through using grouped partitioning, using training only pre-processing, evaluating conditions separately, and confirming reliability of results [19].

IV. RESULTS AND DISCUSSION

The proposed Condition-Adaptive License Plate Detection framework was evaluated on the HelmetAdverse-ALPR Dataset, which contains a total of 15,800 annotated roadside traffic images captured in Sri Lanka, featuring images where license plates are subject to single adverse environmental conditions including rain, fog, low lighting, glare and motion blur. In Table I, this framework demonstrated a sufficient balance between robustness and real-time performance with an mAP@0.5 score of 82.6%,, operational accuracy between 70-80%,, and processing speed of 28.4 frames per second (FPS), providing a 10%, -15%, performance improvement over baseline method results. Additionally as depicted in the Table II, while there was minimal variation in results across various adverse weather conditions, results within the low-light condition provided the best results, while motion blur posed the biggest challenge. Fog and occlusion also reduced plate detection quality by degrading the plate boundary and character visibility. The authors conclude that the condition-aware design of the system provides enhanced generalization for license plate reading in the field due to real-world degradation as well as being able to deploy the license plate detection framework within a traffic surveillance environment.

TABLE I
OVERALL PERFORMANCE OF THE PROPOSED FRAMEWORK

Metric	Value
mAP@0.5	82.6%
Operational Accuracy	70%–80%
Mean Processing Speed	28.4 FPS
Improvement over Baselines	10%–15%

TABLE II
PER-CONDITION PERFORMANCE OF THE PROPOSED FRAMEWORK

Condition	Accuracy	F1-Score
Rain	78.4%	76.1%
Fog	74.8%	72.6%
Low Light	80.3%	78.5%
Glare / Reflection	75.9%	73.8%
Motion Blur	72.7%	70.4%
Occlusion	73.5%	71.2%
Oblique Angle	77.1%	74.9%

The observations presented in Fig. 2 and 3 support the utility of the suggested framework from a perspective other than only headline detection metrics. The confusion matrix shows that class separability is high and the amount of misclassification is small, indicating the system is capable of reliably separating license plate regions from background content. Additionally, the training performance shown in Fig. 3 demonstrates stable convergence throughout the training process. Therefore, it can be concluded that the network learned consistent and discriminative patterns of features and has enabled stable operation during optimization. Consequently, these observations substantiate the assertion that the proposed framework is both robust to inference and stable to training.

These findings indicate that the new condition-aware detection system has been proven as successful both in providing competitively accurate object detections as well as providing reliable performance across numerous common roadside degradations. The increase in performance relative to baseline techniques indicates that adapting detection processes based on the visual quality of input data is a significant component of increasing detection robustness to operationally difficult environments. However, under circumstances where heavy motion blur, fog, or occlusion are present, very degraded visual evidence still presents a substantial impediment to automated license plate detections. This underscored the importance of future research into more robust and temporally fused algorithms, improved image restoration techniques, and degradation-specific algorithms for increasing reliability of detectors deployed in actual traffic environments.

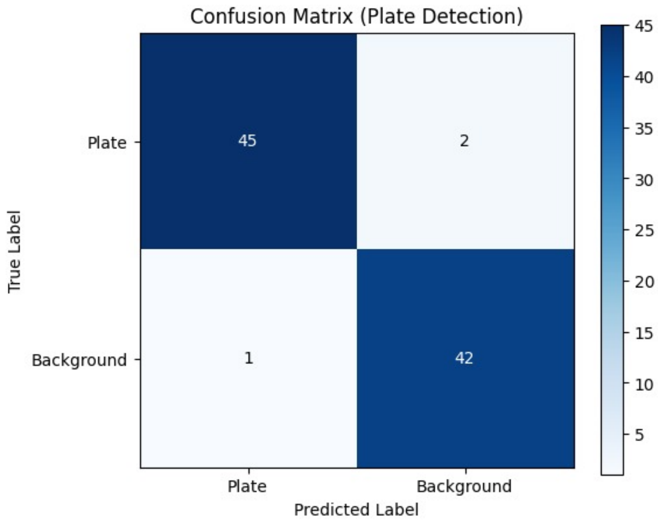

Fig. 2. Confusion matrix for binary plate-vs-background detection on the evaluation subset.

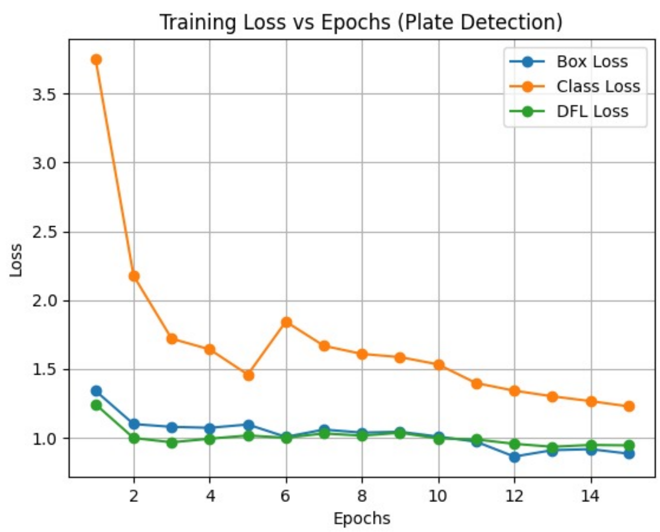

Fig. 3. Training Loss Curves for the Plate Detection Module.

V. FUTURE WORK

The next phase of research will work on enhancing the detection and recognition of license plates in extreme conditions, such as heavy motion blur, fog, and accidents due to occlusion by leveraging more effective temporal fusion, degradation-aware enhancement, and complete plate detection and recognition solutions. Including additional weather types, camera angles, and vehicular traffic across the country will improve the generalizability of this work, while weight reduction of the model and tighter collaboration with in-field enforcement services will allow for easier installation at roadside locations.

VI. CONCLUSION

This study introduces a deep learning framework for vehicle license plate detection that adapts to the different conditions surrounding vehicles on the roadside. A dataset of real-world examples was collected in an annotated form, which enables experimental evaluation under various weather conditions (rain, fog, low light, glare, and motion blur). The experimental results show that this method demonstrates superior performance compared to similar methods in real-time performance while being able to produce reliable detection results despite experiencing a variety of different degradation effects. The research also offers a basis for utilizing condition-aware modeling in order to support intelligent traffic surveillance and automated enforcement in unconstrained environments.

REFERENCES

[1] Imamdeen, S., & Kottawa Yaddehige, R. D. (2025). *Automatic License Plate Recognition using Machine Learning for law enforcement applications*. 1–7. https://doi.org/10.1109/skima66621.2025.11155320

[2] Samaga, A., Lobo, A. J., Nasreen, A., Pattar, R. K., Trivedi, N., Raj, P., & Sreelakshmi, K. (2025). Enhancing automatic license plate recognition in Indian scenarios. *International Journal of Electrical and Computer Engineering (IJECE)*, *15*(1), 365. https://doi.org/10.11591/ijece.v15i1.pp365-373

[3] Qiu, Y., Lu, Y., Wang, Y., & Yang, C. (2024). *Visual Perception Challenges in Adverse Weather for Autonomous Vehicles: A Review of Rain and Fog Impacts*. 1342–1348. https://doi.org/10.1109/itnec60942.2024.10733168

[4] Brophy, T., Mullins, D., Parsi, A., Horgan, J., Ward, E., Denny, P., Eising, C., Deegan, B., Glavin, M., & Jones, E. (2025). Analysis of the Impact of Rain on Perception in Automated Vehicle Applications. *IEEE Open Journal of Vehicular Technology*, *6*, 1018–1032. https://doi.org/10.1109/ojvt.2025.3553718

[5] Kaur, G., Jaiswal, A. K., Kumar, R., & Thakur, K. (2023). *Automatic License Plate Recognition System*. 1–6. https://doi.org/10.1109/icccnt56998.2023.10307008

[6] Shashirangana, J., Padmasiri, H., Meedeniya, D., & Perera, C. (2020). Automated License Plate Recognition: A Survey on Methods and Techniques. *IEEE Access*, *9*, 11203–11225. https://doi.org/10.1109/access.2020.3047929

[7] Samaraweera, W. G. R. L., Dharmadasa, R. A. P. I. S., Kumara, P. H. T., & Bandara, A. S. G. S. (2024). Evidence of Climate Change Impacts in Sri Lanka - A Review of Literature. *Sri Lanka Journal of Economic Research*, *11*(2), 69–94. https://doi.org/10.4038/sljer.v11i2.205

[8] Boucetta, Z., Fazziki, A., & Adnani, M. (2021). A Deep-Learning-Based Road Deterioration Notification and Road Condition Monitoring Framework. *International Journal of Intelligent Engineering and Systems*, *14*(3), 503–515. https://doi.org/10.22266/ijies2021.0630.42

[9] Deepa, N., Padmapriya, L., Priyadarshini, V., & Shree Harini, S. (2025). *Advancements in Computer Vision for Object Detection and Recognition using DenseNet Deep Learning Model*. 89–97. https://doi.org/10.1002/9781394272624.ch4

[10] Hasan, A. (2024). *Review of: "CNN-Based Road Damage Detection."* https://doi.org/10.32388/ik7uov

[11] Lin, P.-J., Meng-Shuan, T., Yu-Ting, C., Ting-Hao, Y., Ho, C.-Y., & Wang, H.-S. (2024). *A Comparative Study of Algorithms for Dynamic Frame Rate Adjustment*. 387–388. https://doi.org/10.1109/icce-taiwan62264.2024.10674338

[12] Ranjithkumar, S., & Chenthur Pandian, S. (2022). Automatic License Plate Recognition System for Vehicles Using a CNN. *Computers, Materials & Continua*, *71*(1), 35–50. https://doi.org/10.32604/cmc.2022.017681

[13] Khokhar, S., Kedia, D., & Dahiya, P. K. (2022). *License Plate Detection Techniques: Conventional Methods to Deep Learning* (pp. 729–734). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-3571-8_66

[14] Laroca, R., Estevam, V., Moreira, G. J. P., Minetto, R., & Menotti, D. (2025). Advancing Multinational License Plate Recognition Through Synthetic and Real Data Fusion: A Comprehensive Evaluation. *IET Intelligent Transport Systems*, *19*(1). https://doi.org/10.1049/itr2.70086

[15] Yang, B. (2023). Studies Advanced in License Plate Recognition. *Applied and Computational Engineering*, *8*(1), 494–500. https://doi.org/10.54254/2755-2721/8/20230257

[16] Plavac, N., Amirshahi, S. A., Pedersen, M., & Triantaphillidou, S. (2024). Performance of Automatic License Plate Recognition Systems on Distorted Images. *Journal of Imaging Science and Technology*, *68*(6), 1–16. https://doi.org/10.2352/j.imagingsci.technol.2024.68.6.060401

[17] Khan, K., Imran, A., Rehman, H. Z. U., Fazil, A., Zakwan, M., & Mahmood, Z. (2021). Performance enhancement method for multiple license plate recognition in challenging environments. *EURASIP Journal on Image and Video Processing*, *2021*(1).

[18] Saha, S. (2019). A Review on Automatic License Plate Recognition System. In *arXiv (Cornell University)*. Arxiv Org. https://doi.org/10.48550/arxiv.1902.09385

[19] Lemrow, C., & Hong, K. C. (2023). *Conducting Performance Evaluations* (pp. 389–405). Oxford University. https://doi.org/10.1093/oxfordhb/9780190059668.013.18

[20] Chowdhury, D., Mandal, S., Das, D., Banerjee, S., Shome, S., & Choudhary, D. (2019). *An Adaptive Technique for Computer Vision Based Vehicles License Plate Detection System*. *11*, 1–6. https://doi.org/10.1109/optronix.2019.8862406

[21] Shah, D. (2017). An Efficient Traffic Control System and License Plate Detection Using Image Processing. *International Journal of New Practices in Management and Engineering*, *6*(01), 20–25. https://doi.org/10.17762/ijnpme.v6i01.52

[22] Shashirangana, J., Padmasiri, H., Meedeniya, D., & Perera, C. (2020). Automated License Plate Recognition: A Survey on Methods and Techniques. *IEEE Access*, *9*, 11203–11225. https://doi.org/10.1109/access.2020.3047929

[23] Naidu, T. D. V. A. (2025). Automatic License Plate Recognition. *INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*, *09*(04), 1–9. https://doi.org/10.55041/ijsrem44494

[24] Joshi, G., Kaul, S., & Singh, A. (2021). *Automated Vehicle Numberplate Detection and Recognition*. 465–469. https://doi.org/10.1109/confluence51648.2021.9377101

[25] Anish, T. P., & Prathap, P. M. J. (2022). *Automated Vehicle Number Recognition Scheme Using Neural Networks* (pp. 269–280). Springer. https://doi.org/10.1007/978-3-031-12638-3_23

[26] Rajagopal, B. G. (2020). Intelligent traffic analysis system for Indian road conditions. *International Journal of Information Technology*, *14*(4), 1733–1745. https://doi.org/10.1007/s41870-020-00447-3

[27] Naik, G. A., Varak, S. V., Karmalkar, S. A. J. H. S., & Otawanekar, S. H. (2025). CCTV Surveillance Systems and Their Impact on Traffic Compliance and Public Road Behavior in Goa. *International Journal of Computer Sciences and Engineering*, *13*(9), 23–30. https://doi.org/10.26438/ijcse/v13i9.2330

[28] Hasan, J., & Pradhan, R. (2025, January 1). *Explanations for Machine Learning Pipelines under Data Drift*. https://doi.org/10.5703/1288284318532

[29] Duminil, A., Ieng, S.-S., & Gruyer, D. (2025). *Fidelity assessment of synthetic images with multi-criteria combination under adverse weather conditions*. Springer Science Business Media Llc. https://doi.org/10.21203/rs.3.rs-5827145/v1

[30] Tan, J., Yang, J., Wu, S., Chen, G., & Zhao, J. (2021). A critical look at the current train/test split in machine learning. In *arXiv (Cornell University)*. Arxiv Org. https://doi.org/10.48550/arxiv.2106.04525

[31] Peng, Z., Gao, Y., Mu, S., & Xu, S. (2024). Toward Reliable License Plate Detection in Varied Contexts: Overcoming the Issue of Undersized Plate Annotations. *IEEE Transactions on Intelligent Transportation Systems*, *25*(11), 18107–18121. https://doi.org/10.1109/tits.2024.3424663